Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/30/2006
PI Name:	Wall, Conrad Ph.D.		
Project Title:	Advanced Techniques for Assessment of Postural and Locomotor Ataxia, Spatial Orientation and Gaze Stability		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Neurovestibular Adaptation Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	cwall@mit.edu	**Fax:**	FY 617-573-4154
PI Organization Type:	UNIVERSITY	**Phone:**	617-573-4160
Organization Name:	Harvard Medical Shcool		
PI Address 1:	Massachusettes Eye and Ear Infirmary		
PI Address 2:	243 Charles Street		
PI Web Page:	http://www.jvl.meei.harvard.edu/		
City:	Boston	**State:**	MA
Zip Code:	02114	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	10/01/2000	**End Date:**	08/31/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC-9-58-NA.002.04		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The overall goals of this project are to develop "countermeasure assessment criteria" to evaluate recovery from perturbations, turning, circular walking and ascending and descending stairs. We also consider countermeasures using a balance prosthesis and dynamic exercises designed to challenge and increase subjects' balance. We will determine the sensitivity of the countermeasure assessment criteria in evaluating effects the prosthesis and the exercises on postural stability and locomotion. Using human subjects, the specific aims of this project are to: 1. Study body and head movements during precise perturbations of gait during continuous straight locomotion. 2. Study body, head and eye movements during continuous straight or circular locomotion on a circular treadmill. 3. Study body, head and eye movements during ascending and descending a staircase. 4. Study body, head and eye movements during standing, linear walking and treadmill walking with a balance prosthesis designed as a countermeasure for vestibular adaptation. 5. Study the effect of dynamic balance exercises for vestibulopathic subjects upon their ability to stand quietly and to recover from mild perturbations		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We have defined some situations that occur in every day locomotion in which astronauts returning from microgravity and in which patients having subtle vestibulopathies are apt to have trouble but which there are no objective measures currently available to quantify their performance. From these, we have increased our understanding of some of the fundamental processes that govern factors like gaze and head position while moving. There is now the potential for developing more meaningful and sensitive tests of balance function that can be applied to astronauts for countermeasure assessment, and to patients with balance anomalies Our project has applied vibrotactile display of body tilt as a potential countermeasure for ataxia in astronauts. We have shown that this technique can be used successfully to decrease vestibulopathic subjects' sway in response to abrupt perturbations while standing.
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)